Design of Low Noise Fractional-N Frequency Synthesizer Using Sigma-Delta Modulation Technique

N. Fatahi and H. Nabovati

Abstract - The fractional-N frequency synthesis technique provides agile switching in narrow channel spacing systems and alleviates phase-locked loop (PLL) design constraints for phase noise and reference spur. A sigma-delta modulator is represented in this work which reduces spur in output frequency spectrum of the fractional-N frequency synthesizers (FS). This technique also improves the performance of the frequency synthesizer. A fractional-N frequency synthesizer is designed by employing a third-order MASH modulator. A fourth order type II PLL with two out of band poles are used to suppress quantization noise of the modulator. The in-band phase noise of -95 dBc/Hz at 10-kHz offset with a spur of less than -96 dBc is achieved with a reference frequency of 8 MHz and a loop bandwidth of 40 kHz.

I. Introduction

Integer-N frequency synthesizers based on a phase locked loop (PLL) are traditionally suffering from limitations in meeting desired specifications due to the fundamental trade-off between the loop bandwidth and the channel spacing. Regarding their high division ratio, meeting the noise requirement using the integer-N synthesizers is also challenging when implemented with CMOS Technology. A fractional-N PLL has benefits compared with the classical integer-N implementations, namely: better phase noise performance, faster lock, and better spur levels. The advantages of fractional-N PLLs come from both a larger loop bandwidth and higher phase-frequency detector (PFD) frequencies. A larger loop bandwidth implies better voltage controlled oscillator (VCO) phase noise suppression and faster lock time.

On the other hand, fractional-N techniques provide wide bandwidth with narrow channel spacing and alleviate PLL design constraints for phase noise and reference spur. The inherent problem of the fractional-N frequency synthesizer is that the periodic operation of the dual-modulus divider produces spurious tones. Several spur reduction techniques have been proposed in the literature [1]-[4]; among them the sigma-delta modulation technique is considered in this work to enhance the overall synthesizer performance.

The objective of this work is to develop a practical frequency synthesis technique for high spectral purity using the sigma-delta modulation method, which is applicable to low-cost wireless transceivers. In this work, the oversampling modulator performance is analyzed by considering practical design aspects in fractional-N frequency synthesis, and a high-order MASH[1] sigma-delta modulator is proposed to enhance the overall performance, and suppress the phase noise and spur.

II. MASH Sigma-Delta Modulator

Figure 1 shows a block diagram of a first-order sigma-delta modulator. It consists of one difference operator, one unit delay element, one discrete integrator, and one 1-bit quantizer whose output is the sum of the input and quantization noise. The relation between the input and the output of a first order sigma-delta modulator is:

$$Y(z) = F_r(z) + (1 - 1/z)E_{q1}(z) \qquad (1)$$

The output signal in Eq. 1 is the sum of two terms; the first is the input signal and the second term is the 1-bit ADC quantization noise which is frequency shaped by a factor of $(1-1/z)$. The quantization noise power density can be computed by passing from the Z transform to the DFT[2]. Assume that $E_1(f)$ and $E_{1,\Sigma\Delta}(f)$ are the quantization noise power density of a 1-bit quantizer and a first-order sigma-delta modulator, respectively. The relation between the two functions is given as follows [5]:

$$E_{1,\Sigma\Delta}(f) = 4\sin^2(\pi f/f_s)E_1(f) \qquad (2)$$

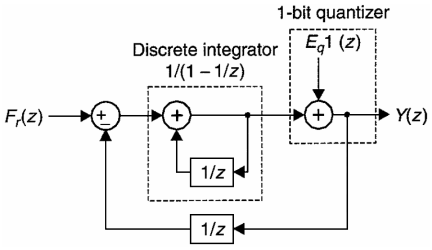


Fig. 1: Block diagram of a first-order Σ-Δ modulator

The 1-bit quantizer output is equal to q if input>q/2 and is equal to zero otherwise. If the input amplitude probability is uniformly distributed in the range [0;q], the quantization noise will follow a uniform distribution too

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[1] Multi stAge noise SHaping
[2] Discrete Fourier Transform

978-1-4244-7201-7/10/$26.00 © 2010 IEEE

between –q/2 and +q/2. Let $P(e_q)$ be the probability density of the quantization error, e_q, therefore the quantization noise power is given by the quantization noise variance as follows:

$$P_1 = \int_{-q/2}^{q/2} e_{q1}{}^2 P(e_q) d(e_q) \quad (3)$$

Assuming that the 1-bit quantization noise density is constant over frequencies from $-f_s/2$ to $+f_s/2$, $E_1(f)$ is given by:

$$E_1(f) = q^2/12 f_s \quad (4)$$

where f_s is the sampling frequency. Consequently $E_{1,\Sigma\Delta}(f)$ is given by:

$$E_{1,\Sigma\Delta}(f) = 4\sin^2(\pi f/f_s)\cdot(q^2/12 f_s) \quad (5)$$

The total quantization noise of the first order Σ-Δ modulator can be calculated by integrating Eq. 2 over the frequency range, therefore:

$$P_{1,\Sigma\Delta} = 2P_1 \quad (6)$$

The total noise of the first order Σ-Δ modulator is twice of that the 1-bit quantizer, but its density shifted towards higher frequencies. The noise shaping property expressed by Eq. 1 suggests a method to combine more Σ-Δ modulators in order to further reduce the low frequency quantization noise power density. Figure 2 shows the block diagram of this multi-stage cascaded modulator, which is often called the MASH modulator for a third order modulator. Output signal of modulator $Y(z)$ is as follows:

$$Y(z) = F_r(z) + (1-1/z)^3 E_{q3}(z) \quad (7)$$

where $E_{q3}(z)$ is the quantization noise of third stage. Generalizing the procedure applied for the third-order Σ-Δ modulator, the M-order Σ-Δ modulator quantization noise power density is given by Eq. 8 [5].

$$E_{M,\Sigma\Delta}(f) = (q^2/12 f_s)\cdot\{2\sin(\pi f/f_s)\}^{2M} \quad (8)$$

III. Design Consideration

An ideal frequency synthesizer generates a stable signal, which is ideally a single tone in the frequency domain. In reality, the signal is not pure, and unwanted pieces of information are added to the signal in two ways: random or deterministic signals. Phase noise and spurious tones often limit the overall synthesizer performance. The noise from the synthesizer and the spurious tones can be reduced by narrowing the PLL bandwidth, but the narrow-band PLL suffers from long settling time. Consequently, there is a trade-off to determine the synthesizer performance in terms of the phase noise, the spurious tone, and the settling time.

Assuming the uniform quantization error, the error power is to be 1/12 where the minimum step size of the quantizer is set to be 1 due to the integer-divider quantization. This noise power is distributed over a bandwidth of the phase detector frequency; and the frequency fluctuation in the z-domain $S_v(z)$ of noise transfer function of $H_n(z)$ is given by [6]:

$$S_v(z) = \left|H_n(z) f_{ref}\right|^2 \cdot (1/12 f_{ref}) \quad (9)$$

where, f_{ref} is reference frequency. Since the phase fluctuation $S_\phi(z)$ is the integration of the frequency fluctuation, it is calculated as follows:

$$S_\phi(z) = \left\{2\pi/\left(\left|1-z^{-1}\right| f_{ref}\right)\right\}^2 \cdot \left(\left|H_n(z)\right|^2 f_{ref}/12\right) \quad (10)$$

If $S_\phi(z)$ is a two-sided power spectral density (PSD), then the single-sided PSD $L(z)$ is the same as $S_\phi(z)$. According to Eq. 7, the noise transfer function for the third-order MASH modulator is $\left|1-z^{-1}\right|^3$, where the $L(z)$ is given by:

$$L(z) = \left((2\pi)^2/12 f_{ref}\right)\cdot\left|1-z^{-1}\right|^4 \quad (11)$$

Converting to the frequency domain and generalizing to any modulator order, we can obtain:

$$L(f) = \left((2\pi)^2/12 f_{ref}\right)\cdot\left\{2\sin(\pi f/f_{ref})\right\}^{2(M-1)} \quad (12)$$

Phase noise is the randomness of the frequency instability; and in the RF frequency synthesizers, it is considered to be one of the most important specifications. In the frequency synthesizers, the phase noise within the loop bandwidth mainly determines the integrated phase error [7]. Accordingly, wide-band synthesizers should achieve low in-band phase noise to maintain the same integrated phase error.

By interpreting the noise as a normalized signal within 1-Hz bandwidth, the phase noise can be analyzed by employing narrow-band FM theory [8]-[10]. The integrated phase noise is determined:

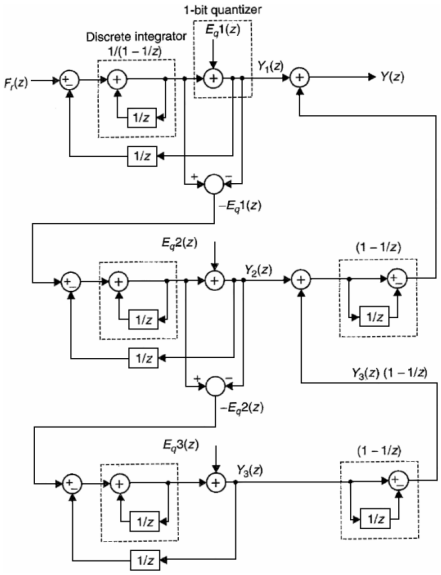


Fig. 2: Third-order Σ-Δ MASH modulator

$$\Delta\theta_n = \sqrt{\int_a^b 2L(f_m)df_m} \tag{13}$$

While the phase noise represents the randomness of the frequency stability, the spurious tone is set by the deterministic behavior. Since the spur generation is considered the systematic offset in the PLL, it is relatively easy to improve the spur performance up to a certain level compared to improving the phase noise performance. As the VCO is naturally a voltage-to-frequency converter, the spurs in a PLL result from the FM modulation by the VCO. One source of the reference spur is the leakage current caused by the charge pump itself or by any leakage current on the board. The phase offset, ϕ_ε, due to the leakage current, I_{Leak}, with the charge pump current, I_{CP}, is $\phi_\varepsilon = (2\pi I_{Leak})/I_{CP}$.

For an oversampling Σ-Δ modulator, the decimation filter is necessary to enhance the overall dynamic range. In frequency synthesis, this filter cannot be used since the Σ-Δ modulator has to operate at the phase detector frequency for the best performance, as explained previously. A PLL, however, will do a similar job since its transfer function suppresses the high-frequency noise as a low-pass filter. Since a third-order Σ-Δ modulator is used, a type II, fourth-order PLL having two additional out-of-band poles is designed to filter out the quantization noise of the modulator at high frequencies. The amount of the spur, p_{spur}, (dBc) in the fourth-order PLL is given by:

$$\begin{aligned} p_{spur} &= 20Log\left(f_{BW}N\phi_\varepsilon/f_{ref}\right) \\ &\quad - 20Log\left(f_{ref}/f_{p1}\right) - 20Log\left(f_{ref}/f_{p2}\right) \end{aligned} \tag{14}$$

where R is the resistor in the loop filter, K_{VCO} is the VCO gain, f_{ref} is the reference frequency for the PFD, f_{p1} and f_{p2} are the pole frequencies in the loop filter and the loop bandwidth, f_{BW}, with a division ratio of N is approximated as $f_{BW} \approx (I_{CP}RK_{VCO})/2\pi N$.

IV. Fractional-N Frequency Synthesizer Design and Simulation

The proposed third-order MASH Σ-Δ modulators for the fractional-N synthesizer does not use the decimation filter to suppress the high-frequency noise in the digital domain since the PLL with integer-N dividers does not allow an intermediate level between N and N+1. Therefore, the clock frequency of the oversampling modulator must be the same as the phase detector frequency in order to not increase the quantization noise. However, the PLL acts as a low-pass filter to the quantization noise, and the noise-shaped oversampling technique can be realized even though the oversampling modulator is working at the PFD. Since a third-order modulator is used in the proposed design, a type II fourth-order PLL with two additional out-of-band poles is used to filter out the quantization noise at high frequencies.

Since the MASH modulator generates the multi-bit output, a multi-modulus divider is used. In this work, a tri-state charge pump with the PFD is employed to avoid using an active filter. In order to make a type II PLL possible with the passive loop filter a tri-state charge pump is used. The lowpass filter consists of a first order lowpass filter followed by two RC filters. The cut-off frequency of the first order low pass filter is determined by $\omega_{LPF} = 1/RC$. Consequently, selection of capacitor with nominal value of 330pF and $f_{ref}/15$ cut-off frequency results in a 5.65 kΩ of resistor in the loop filter. A main constraint for the two out of band poles is that the parasitic pole frequencies should be at least one order of magnitude higher than the PLL bandwidth in order to allow proper pole placement of the dominant poles of the PLL. According to Eq. 14, when the two out of band poles, f_{p1}, f_{p2} are placed at $0.1f_{ref}$ and $0.2f_{ref}$, the spur suppression is calculated to be about -34 dBc/Hz.

The first term in Eq. 14 is the spur power without suppression. According to this term, the magnitude of a spur depends on the VCO gain, K_{VCO}, and the magnitude of a charge pump current, I_{CP}; therefore, a low K_{VCO} should be used to reduce the spur power. As a result, the VCO with a sensitivity K_{VCO} of 50 MHz/V is used and the charge-pump output current, I_{CP}, is set to be 100 μA. In submicron CMOS, a leakage current as high as 1 nA can be easily observed, therefore the magnitude of the leakage current is considered to be 10 nA in the current design.

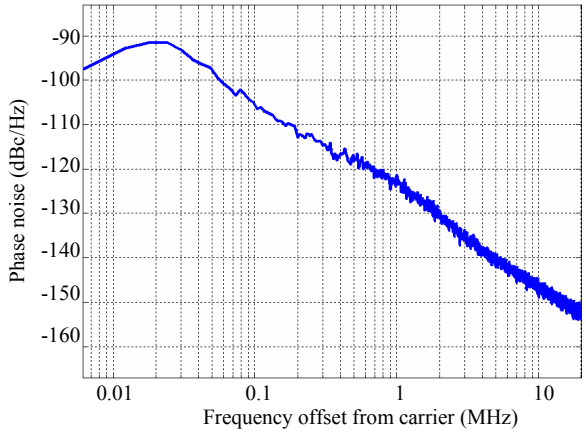


Fig. 3: Output phase noise of fractional-N FS

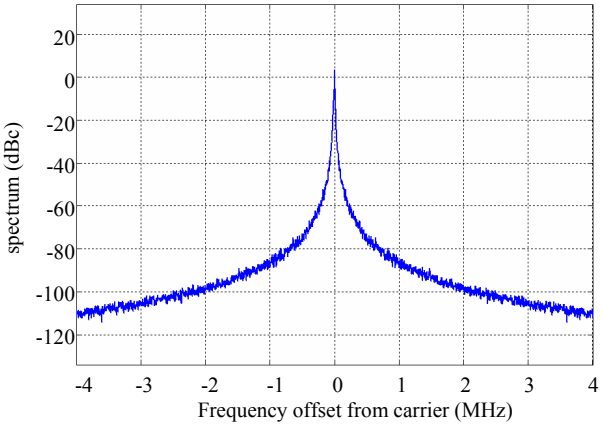


Fig. 4: Output spectrum of fractional-N FS

For a 900 MHz output with a 8 MHz phase detector frequency, the division ratio N is equal to 112.5; the bandwidth frequency is approximately 40 kHz. Figure 3 shows the output phase noise of fractional-N frequency synthesizer at 900 MHz with the third-order MASH modulator. As shown in Fig. 3, the phase noise value at 10 kHz offset is about -95 dBc/Hz.

Figure 4 shows the simulated output spectrum at 900 MHz with the third-order MASH modulator. As expected, the third-order modulator case shows less out-of-band noise. With the 8 MHz reference frequency, spur is -96 dBc at about 4 MHz offset. The out-of-band spur is suppressed substantially by the loop filter as the offset frequency increases. Table 1 shows the performance comparison with the previously published works. The phase noise and spur level with [11] are respectively -92 dBc/Hz and -95 dBc. Comparing phase noise and spur level with [11], the in-band phase noise at 10 kHz improved 3 dBc/Hz; and the spur level is also suppressed by 1 dBc and more.

V. Conclusion

In order to achieve a better in-band phase noise and lower spurs, a new fractional-N frequency synthesizer with a third-order MASH modulator was presented. A type II fourth-order PLL with two additional out-of-band poles is used to filter out the quantization noise at high frequencies. The synthesizer achieves a close-in phase noise of -95 dBc/Hz at an offset frequency of 10 kHz, while the spurious tones are at a level of -96 dBc.

Table 1. Performance comparison with existing arts

Ref.	Mod.	Output freq.	Reference freq.	Phase noise	spur
W. Rhee, [11]	3rd-order 3-bit	900 MHz	40 KHz	-92 dBc/Hz	-95 dBc
M. Perrott, [12]	2nd-order MASH	1.8 GHz	84 KHz	-74 dBc/Hz	-60 dBc
A. Ahola, [13]	3rd-order 8-bit	1760 MHz	50 KHz	-81 dBc/Hz	-85 dBc
This work	3rd-order MASH	900 MHz	40 KHz	-95 dBc/Hz	-96 dBc

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